

Gauging Hourly Economic Activity in Your Neighborhood

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¹**Disclaimer:** The views expressed here are my own and do not necessarily represent the views of the Federal Reserve Bank of Philadelphia or the Federal Reserve System.

This paper ...

Monitors economic activity in San Diego at a granular level.

The paper combines three modern features of real-time econometric monitoring:

- Higher frequency: hourly observations.
- Finer geography: 62 ZIP codes within one county.
- Broader data: traffic, electricity, and spending.

The measure has academic and policy uses beyond real-time monitoring:

- Across time: event studies and spending responses to transfers.
- Across space: policy comparisons near geographic boundaries.

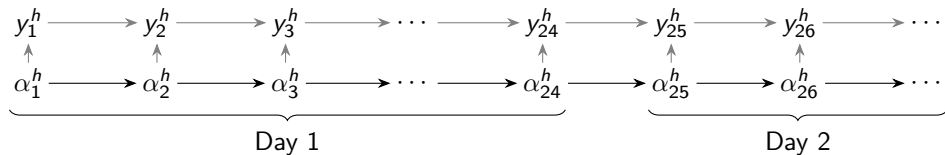
Contributions

1. New evidence on hourly economic activity across locations.
2. A model that combines mixed frequencies and spatial scales, linking alternative data to payroll and GDP.
3. A new filtering algorithm for mixed-frequency dynamic factor models.

My focus today is the econometric contribution: the Cascade Kalman filter.

Comments on the Cascade Kalman Filter

A simpler framework



Hourly model

Hourly measurement

$$y_t^h = \alpha_t^h + \epsilon_t^h$$

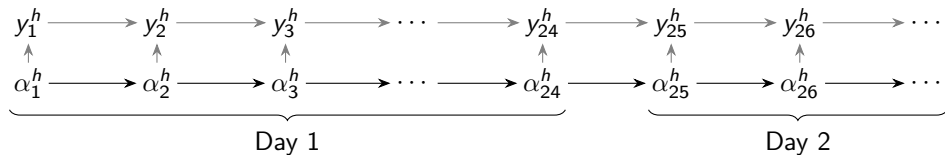
Latent activity

$$\alpha_{t+1}^h = T\alpha_t^h + \eta_{t+1}^h$$

Innovations

$$\epsilon_t^h \sim N(0, H^h), \quad \eta_{t+1}^h \sim N(0, Q)$$

A simpler framework



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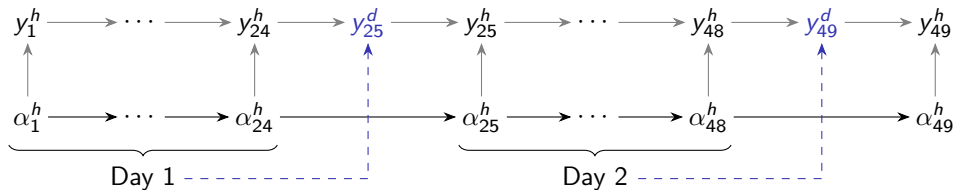
Learning from hourly data

$$E(\alpha_t^h | \mathcal{Y}_{t-1}^h, y_t^h) = b_0 + b_1 y_t^h$$

$$b_1 = \frac{\text{Cov}(\alpha_t^h, y_t^h | \mathcal{Y}_{t-1}^h)}{\text{Var}(y_t^h | \mathcal{Y}_{t-1}^h)}.$$

This is a conditional linear projection.

The mixed-frequency problem



A daily signal arrives after hour 24 and before hour 25.

$$y_{25}^d = \alpha_{25}^d + \epsilon_{25}^d, \quad \epsilon_{25}^d \sim N(0, H^d).$$

Daily latent activity α_{25}^d aggregates the hourly states:

$$\alpha_{25}^d = \alpha_1^h + \alpha_2^h + \alpha_3^h + \dots + \alpha_{24}^h.$$

Learning from daily data

Daily model

Daily measurement

$$y_{25}^d = \alpha_{25}^d + \epsilon_{25}^d,$$

$$\epsilon_{25}^d \sim N(0, H^d).$$

Aggregation

$$\alpha_{25}^d = \sum_{j=1}^{24} \alpha_j^h.$$

Learning from daily data

$$E(\alpha_{25}^h \mid \mathcal{Y}_{24}^h, y_{25}^d) = \beta_0^{\text{KF}} + \beta_1^{\text{KF}} y_{25}^d.$$

$$\beta_1^{\text{KF}} = \frac{\text{Cov}(\alpha_{25}^h, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)}.$$

$$\text{Cov}(\alpha_{25}^h, y_{25}^d \mid \mathcal{Y}_{24}^h) = T \sum_{j=1}^{24} \text{Cov}(\alpha_{24}^h, \alpha_j^h \mid \mathcal{Y}_{24}^h).$$

$$\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h) = \sum_{i=1}^{24} \sum_{j=1}^{24} \text{Cov}(\alpha_i^h, \alpha_j^h \mid \mathcal{Y}_{24}^h) + H^d.$$

Full history: 24 hourly states for a day, 8,760 for a year. Can we retain fewer?

The Cascade Kalman filter

Use y_{25}^d to update the state at hour 25 before observing y_{25}^h .

$$E(\alpha_{25}^h \mid \mathcal{Y}_{24}^h, y_{25}^d) = \beta_0 + \beta_1^{\text{KF}} y_{25}^d.$$

Key idea: Use the daily variance as the numerator of the hourly gain:

$$\beta_1^{\text{Cascade}} = \underbrace{\frac{\text{Cov}(\alpha_{25}^d, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)}}_{\text{LP of } \alpha_{25}^d \text{ on } y_{25}^d} \quad \text{instead of} \quad \beta_1^{\text{KF}} = \underbrace{\frac{\text{Cov}(\alpha_{25}^h, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)}}_{\text{LP of } \alpha_{25}^h \text{ on } y_{25}^d}.$$

The covariance on the left is much easier to compute because $y_{25}^d = \alpha_{25}^d + \epsilon_{25}^d$.

The Cascade Kalman filter (continued)

To implement this idea, we need to calculate the daily variance.

$$\beta_1^{\text{Cascade}} = \frac{\text{Cov}(\alpha_{25}^d, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)} = \frac{\text{Var}(\alpha_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(\alpha_{25}^d \mid \mathcal{Y}_{24}^h) + H^d}$$

Approximate the daily variance using a single filtered hourly variance:

$$\begin{aligned}\text{Var}(\alpha_{25}^d \mid \mathcal{Y}_{24}^h) &= \sum_{i=1}^{24} \sum_{j=1}^{24} \text{Cov}(\alpha_i^h, \alpha_j^h \mid \mathcal{Y}_{24}^h) \\ &\approx 24 \text{Var}(\alpha_{24}^h \mid \mathcal{Y}_{24}^h)\end{aligned}$$

The Cascade Kalman filter (continued)

Assign the daily posterior variance to the hourly state.

Cascade assigns

$$\text{Var}_{\text{Cascade}}(\alpha_{25}^h | \mathcal{Y}_{24}^h, y_{25}^d) = \underbrace{\text{Var}(\alpha_{25}^d | \mathcal{Y}_{24}^h)}_{\text{daily prior variance}} - \frac{\text{Cov}(\alpha_{25}^d, y_{25}^d | \mathcal{Y}_{24}^h)^2}{\text{Var}(y_{25}^d | \mathcal{Y}_{24}^h)}.$$

Exact hourly posterior variance

$$\text{Var}(\alpha_{25}^h | \mathcal{Y}_{24}^h, y_{25}^d) = \underbrace{\text{Var}(\alpha_{25}^h | \mathcal{Y}_{24}^h)}_{\text{hourly prior variance}} - \frac{\text{Cov}(\alpha_{25}^h, y_{25}^d | \mathcal{Y}_{24}^h)^2}{\text{Var}(y_{25}^d | \mathcal{Y}_{24}^h)}.$$

Exact updating also requires the hourly–daily covariance.

Cascading across frequencies

Suppose data at all four frequencies arrive at hour t .



- Recompute each gain using the variance passed down.
- Pass the updated estimate and variance to the next frequency in the sequence.
- Process the hourly signal last, then predict the state for the next hour.

Information cascades down the frequency hierarchy, giving the filter its name.

Cascade is an approximation

Benefit: storage and computation

Full annual history: 8,760 states.

Cascade avoids storing the full $8,760 \times 8,760$ covariance matrix.

Cost: approximation

$$\underbrace{\beta_1^{\text{KF}}}_{\text{exact}} = \frac{\text{Cov}(\alpha_{25}^h, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)}$$
$$= \underbrace{\frac{\text{Var}(\alpha_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(y_{25}^d \mid \mathcal{Y}_{24}^h)}}_{=\beta_1^{\text{Cascade}}} \times \underbrace{\frac{\text{Cov}(\alpha_{25}^h, y_{25}^d \mid \mathcal{Y}_{24}^h)}{\text{Var}(\alpha_{25}^d \mid \mathcal{Y}_{24}^h)}}_{=\text{Bias}}$$

An exact alternative: the accumulator

The daily release at hour 25 measures the preceding 24 hours.

$$y_{25}^d = \sum_{j=1}^{24} \alpha_j^h + \epsilon_{25}^d, \quad \epsilon_{25}^d \sim N(0, H^d).$$

Accumulate hourly states through $t - 1$, starting with $\bar{\alpha}_1^d = 0$:

$$\alpha_t^h = T \alpha_{t-1}^h + \eta_t^h, \quad \eta_t^h \sim N(0, Q),$$
$$\bar{\alpha}_t^d = \begin{cases} \alpha_{t-1}^h & \text{if } t - 1 \text{ starts a day,} \\ \bar{\alpha}_{t-1}^d + \alpha_{t-1}^h & \text{otherwise.} \end{cases}$$

$$\bar{\alpha}_{25}^d = \sum_{j=1}^{24} \alpha_j^h, \quad \text{then} \quad \bar{\alpha}_{26}^d = \alpha_{25}^h.$$

Harvey (1989); Aruoba, Diebold, and Scotti (2009).

An exact alternative: the accumulator

Measurement equations

$$y_t^h = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha_t^h \\ \bar{\alpha}_t^d \end{bmatrix} + \epsilon_t^h, \quad y_{25}^d = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_{25}^h \\ \bar{\alpha}_{25}^d \end{bmatrix} + \epsilon_{25}^d.$$

State transition

$$\begin{bmatrix} \alpha_t^h \\ \bar{\alpha}_t^d \end{bmatrix} = \begin{bmatrix} T & 0 \\ 1 & d_t \end{bmatrix} \begin{bmatrix} \alpha_{t-1}^h \\ \bar{\alpha}_{t-1}^d \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \eta_t^h, \quad \eta_t^h \sim N(0, Q).$$

$d_t = 0$ if $t - 1$ starts a day; $d_t = 1$ otherwise.

- Exact filtering and smoothing with a fixed state dimension.
- Other frequencies work similarly; delayed releases may need additional states.

Simulation: Cascade versus exact

We follow the authors' empirical structure with linear temporal aggregation.

- One pooled county; 200 paths; 26,280 hours per path, 2021–2023.
- Seven series: four hourly, one daily (spending), one monthly (payroll), and one annual (GDP).
- Local linear trend model (level and slope), with known calendar effects.
- Filtering performance:

Filter	State update	RMSE	Time (s/path)
Cascade KF	Authors' update rules	0.26	0.06
Exact KF	Accumulator	0.14	0.11

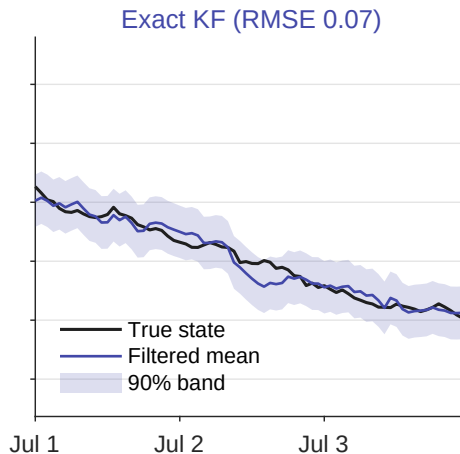
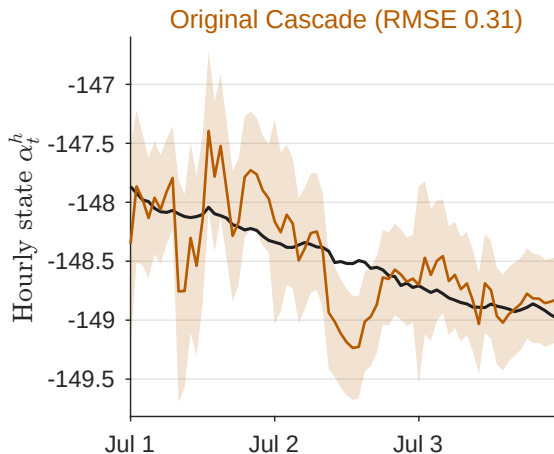
Simulation: Cascade versus exact

I asked Claude Code (Opus 5.5) to optimize my MATLAB code... (Costed about \$5 ...)

Filter	State update	RMSE	Time (s/path)	
			MATLAB	Optimized C
Cascade KF	Authors' update rules	0.26	0.06	0.00022
Exact KF	Accumulator	0.14	0.11	0.00024

True state and filtered estimates

One simulated path, July 1–3, 2023.



Comments on nonlinear aggregation

Nonlinear temporal aggregation

The authors model latent activity on a log scale:

$$\alpha_t^h = T\alpha_{t-1}^h + \eta_t^h$$

Temporal aggregation:

$$\alpha_t^d = \log \left(\sum_{j=1}^{24} \exp \left(\alpha_{t-j}^h \right) \right)$$

Simple example, revisited

This leads to a nonlinear state-space model:

$$y_t^d = \log \left(\sum_{j=1}^{24} \exp \left(\alpha_{t-j}^h \right) \right) + \epsilon_t^d, \quad \epsilon_t^d \sim N(0, H^d)$$
$$\alpha_t^h = T \alpha_{t-1}^h + \eta_t^h, \quad \eta_t^h \sim N(0, Q)$$

The authors use a first-order Taylor expansion to approximate the aggregate variance.

- Conditional moments and updates are approximate. In what sense does Cascade fully enforce temporal aggregation?
- Benchmark against nonlinear alternatives, such as the unscented Kalman filter.

ADS implementation: growth to growth

An hourly-to-daily version of the ADS growth specification:

$$\underbrace{\Delta_d y_t^d}_{\text{daily log growth}} = c + \beta \sum_{j=1}^{24} \alpha_{t-j}^h + \text{lag terms} + \epsilon_t^d.$$

Here $\Delta_d y_t^d = y_t^d - y_{t-24}^d$, and α_t^h is an hourly economic activity factor.

- Flow aggregation is also approximate.
- Exact Kalman filtering for the specified linear Gaussian model.

Philadelphia Fed, Updates on ADS Index Calculation, December 2008 update.

Conclusion

Modern elements of real-time economic activity monitoring:

- Higher frequency: hourly observations.
- Finer geography: 62 ZIP codes within one county.
- Broader data: traffic, electricity, and spending.

Cascade KF: a new filtering method for mixed-frequency dynamic factor models.

My main suggestion is to benchmark the proposed method and model against alternatives:

- Exact Kalman filter with accumulator states.
- Unscented Kalman filter for nonlinear aggregation.
- ADS-style growth model with approximate flow aggregation.